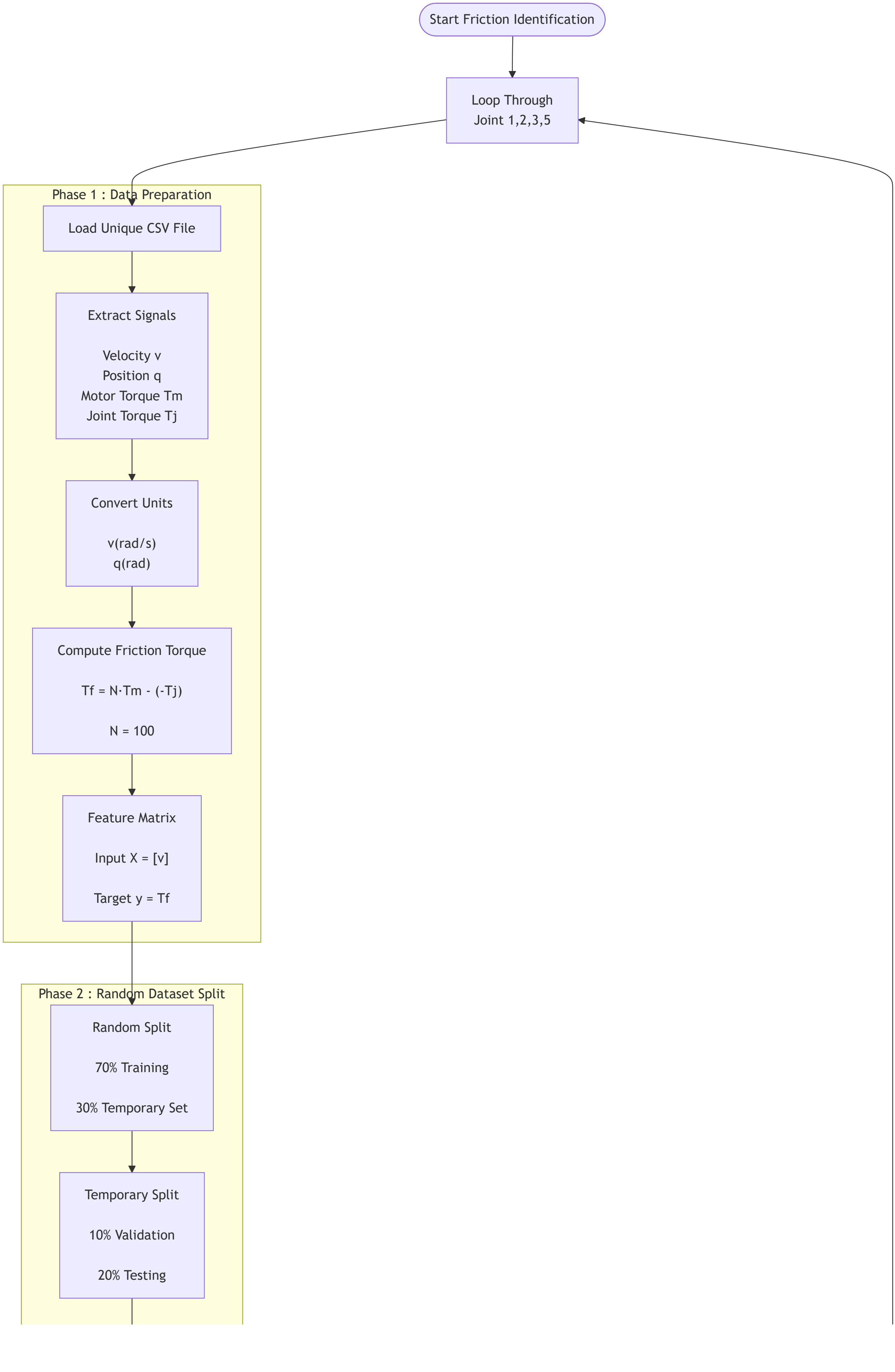
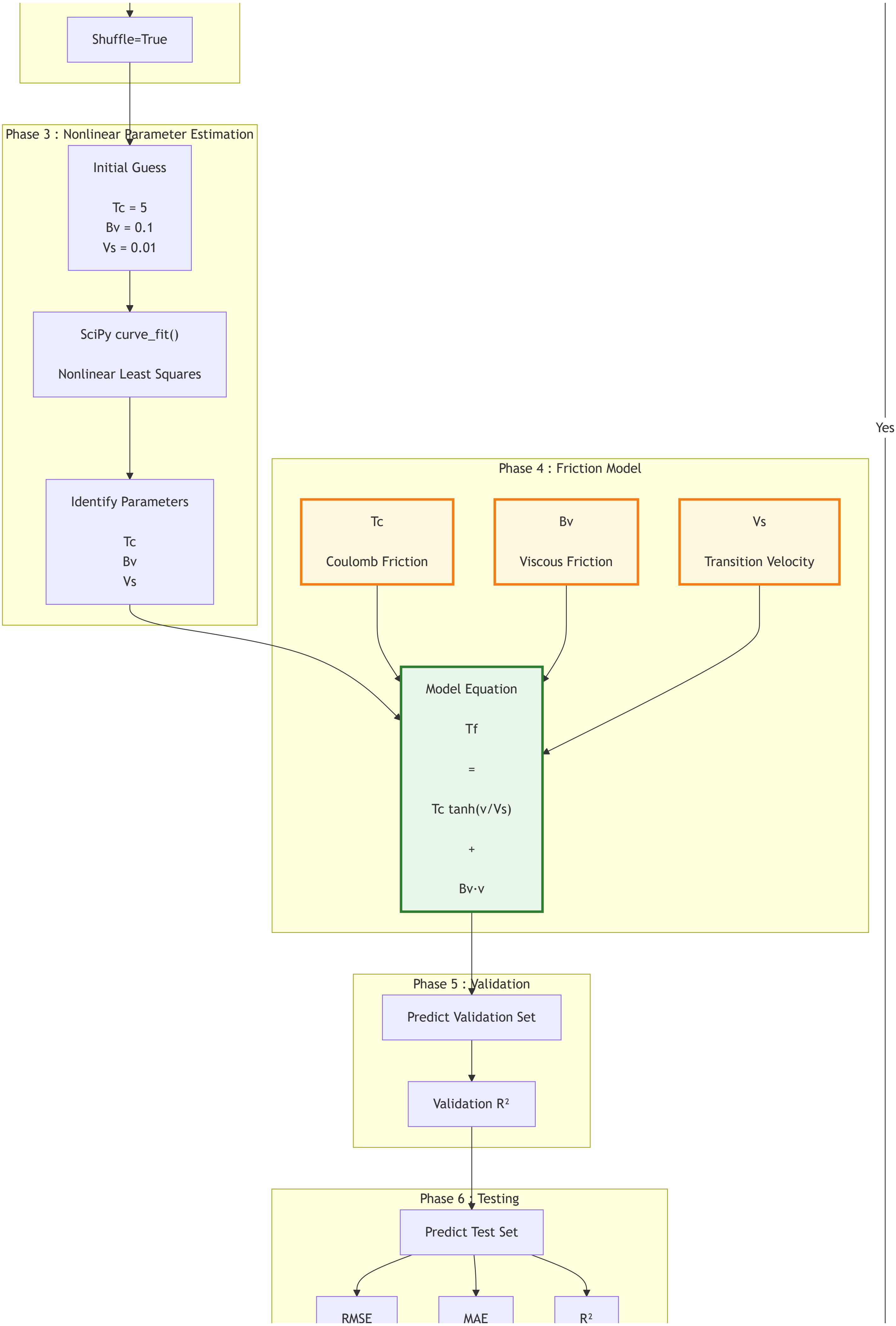
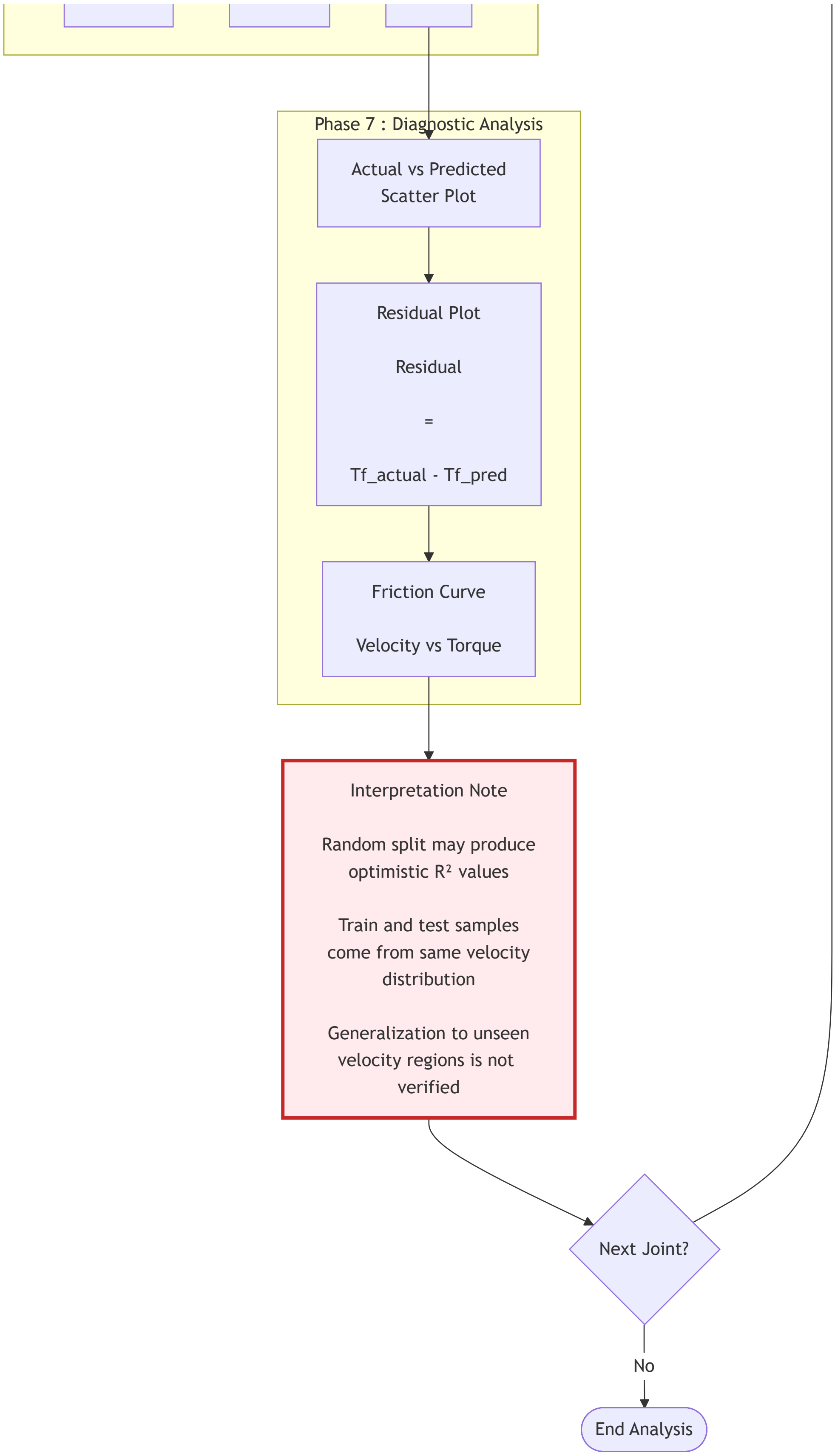

Start Friction Identification
Loop Through
Joint 1,2,3,5
Phase 1 : Data Preparation
Load Unique CSV File
Extract Signals
Velocity v
Position q
Motor Torque Tm
Joint Torque Tj
Convert Units
v(rad/s)
q(rad)
Compute Friction Torque
Tf = N·Tm - (-Tj)
N = 100
Feature Matrix
Input X = [v]
Target y = Tf
Phase 2 : Random Dataset Split
Random Split
70% Training
30% Temporary Set
Temporary Split
10% Validation
20% Testing


Shuffle=True
Phase 3 : Nonlinear Parameter Estimation
Initial Guess
Tc = 5
Bv = 0.1
Vs = 0.01
SciPy curve_fit()
Nonlinear Least Squares
Identify Parameters
Tc
Bv
Vs
Yes
Phase 4 : Friction Model
Tc
Coulomb Friction
Bv
Viscous Friction
Vs
Transition Velocity
Model Equation
Tf
=
Tc tanh(v/Vs)
+
Bv·v
Phase 5 : Validation
Predict Validation Set
Validation R²
Phase 6 : Testing
Predict Test Set
RMSE
MAE
R²


Phase 7 : Diagnostic Analysis
Actual vs Predicted Scatter Plot
Residual Plot
Residual
=
Tf_actual - Tf_pred
Friction Curve
Velocity vs Torque
Interpretation Note
Random split may produce optimistic R² values
Train and test samples come from same velocity distribution
Generalization to unseen velocity regions is not verified
Next Joint?
No
End Analysis